

① Safety-Critical Personalization

Clinical Need

 **Allergy information** (severe) ⚠️

 **Drug contraindication**
ACE inhibitors → risk ⚠️

 **Past history**
chronic gastritis ⚠️

Existing Benchmarks

 Favorite food
Pizza

 Daily preference
Likes football, jazz

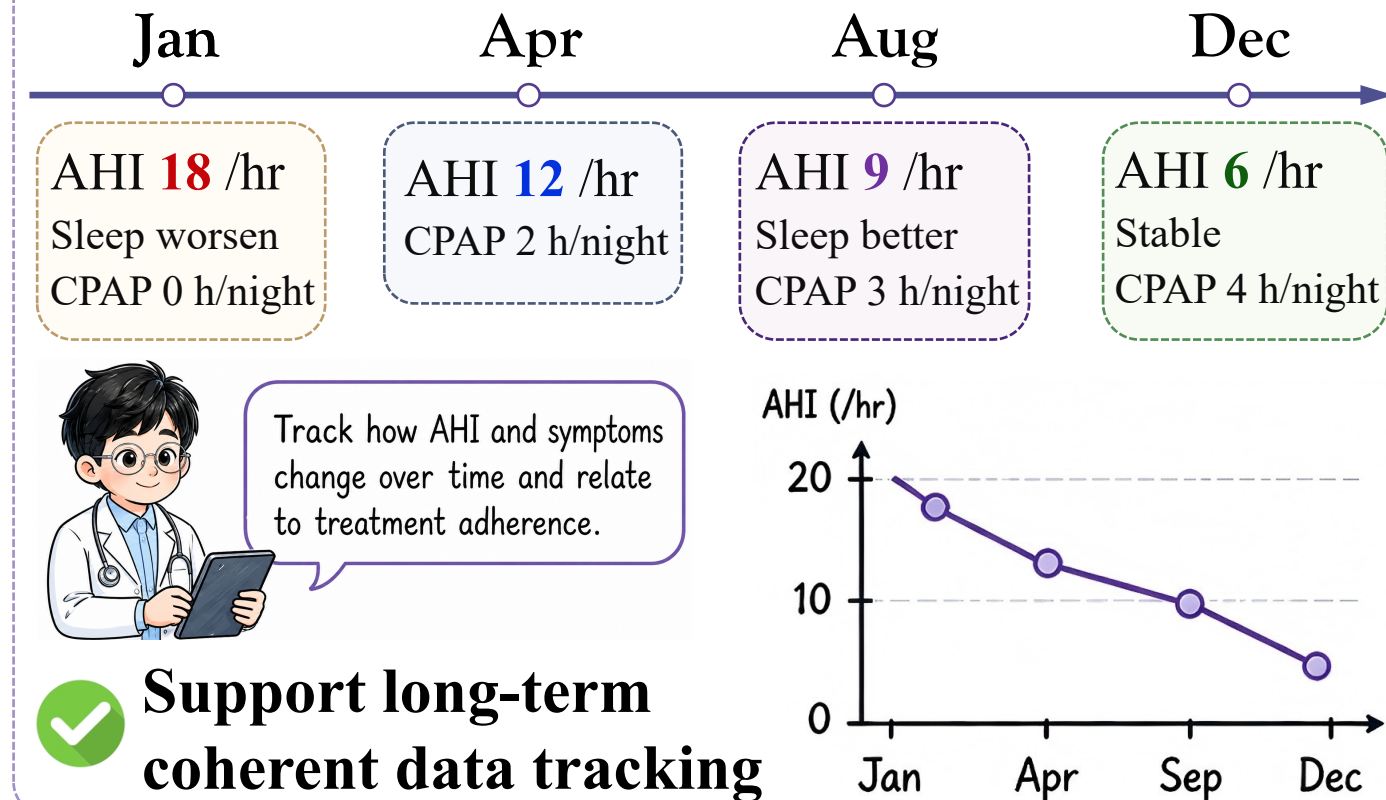
 Persona fact
Walk the pet dog...

VS.

Sensitive information must be **firmly remembered and strongly secured**. But in general domains, there is generally no distinction between strong and weak.

② Longitudinal Clinical Progression

Temporal States



Existing Benchmarks

Short-term Data

AHI: 9 /hr
CPAP: 3 h/night
Symptoms: better

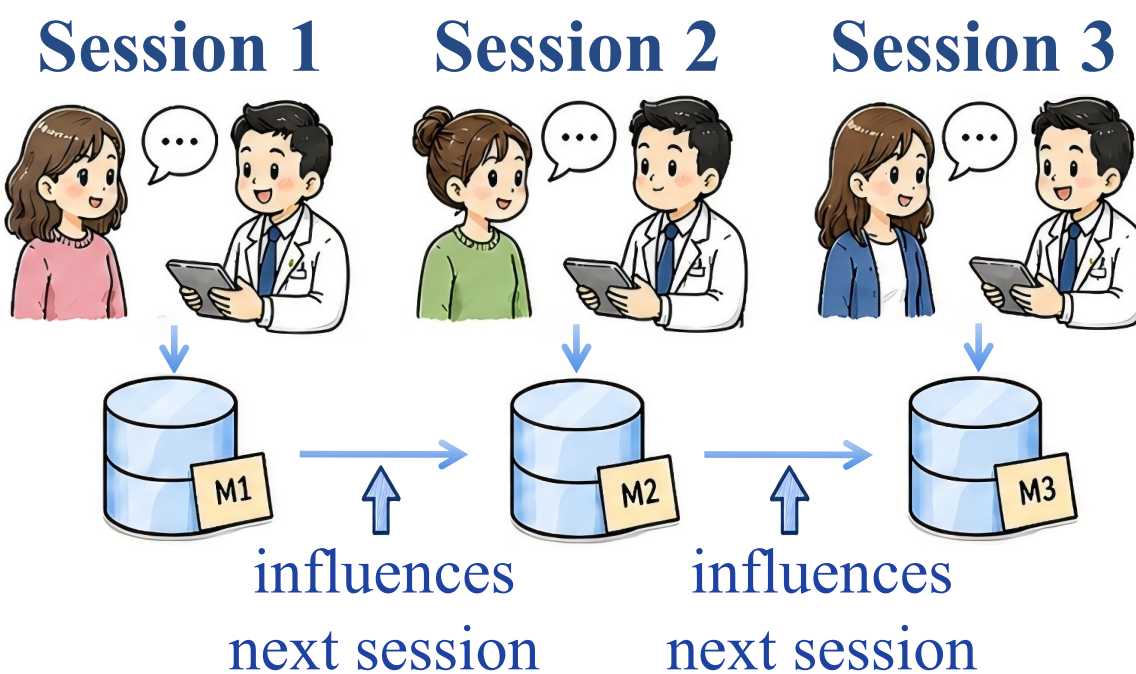
Overwrite old state



Unable to evaluate long-term state changes.

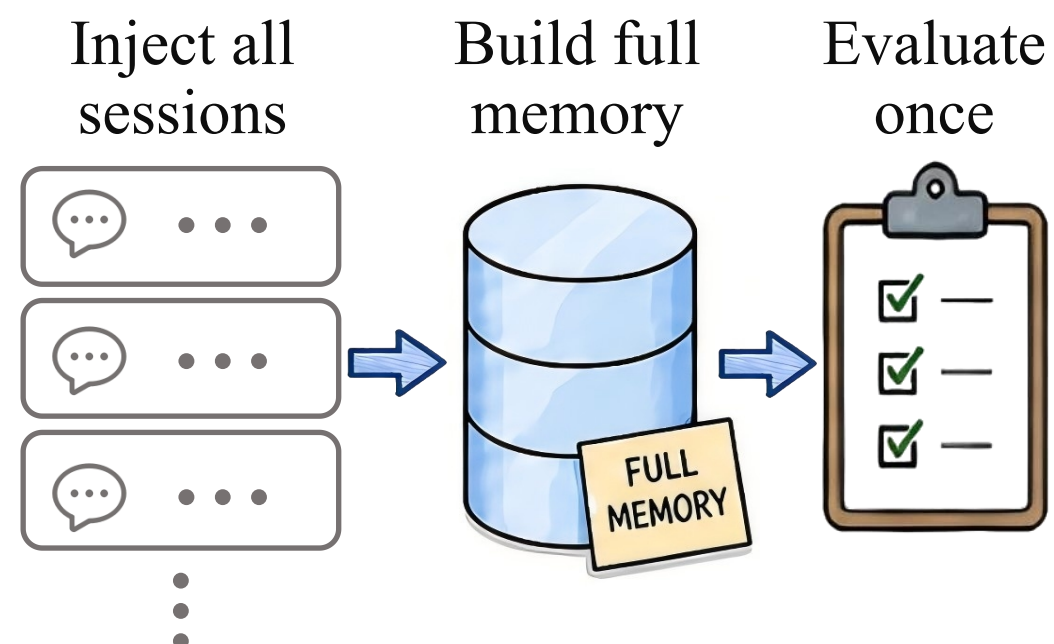
③ Streaming Memory Integration

Real Workflow



In real scenarios, memory is **accumulated online** and **influences future sessions**.

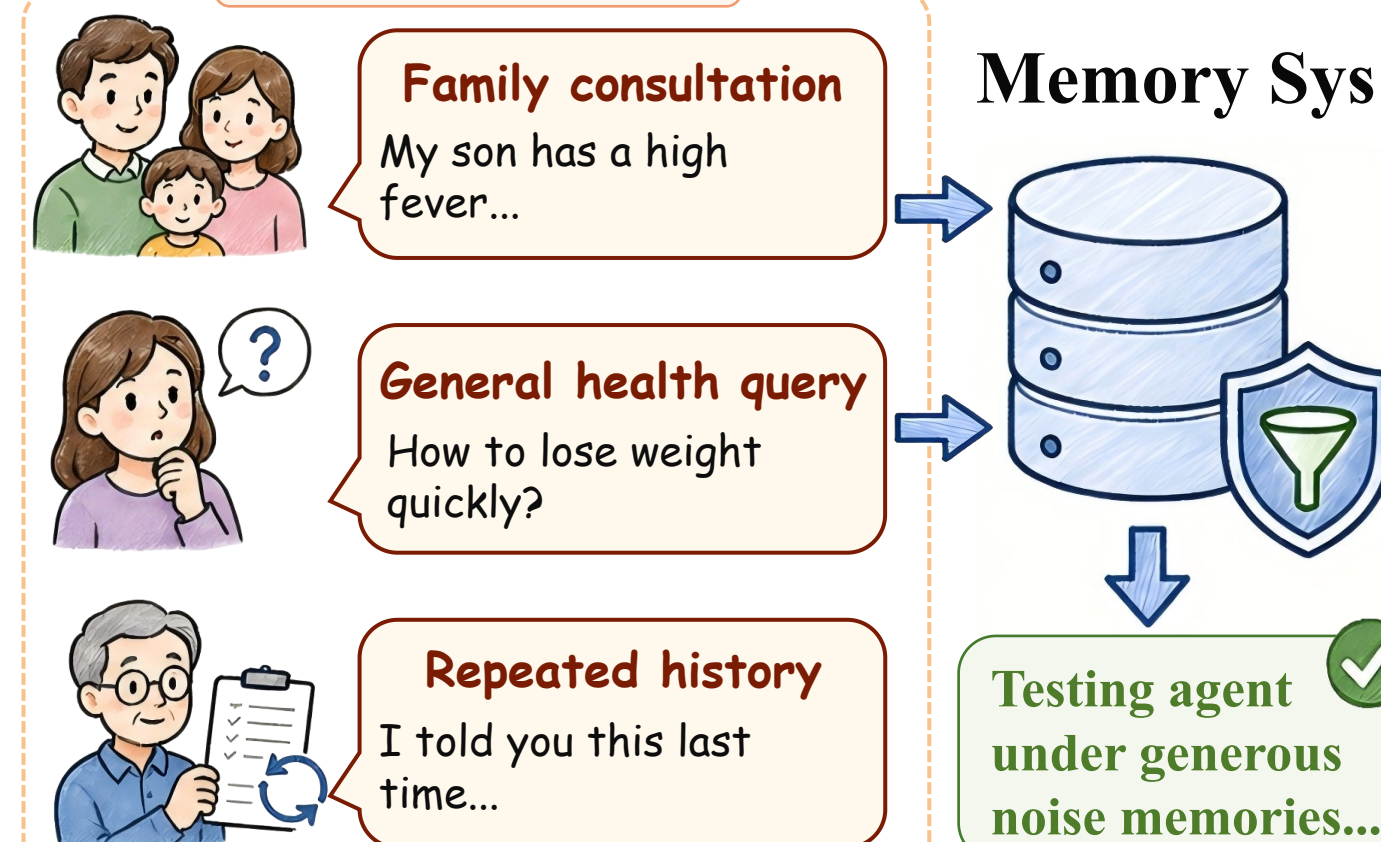
Existing Benchmarks



offline one-shot assessment, **mismatched with real deployment**.

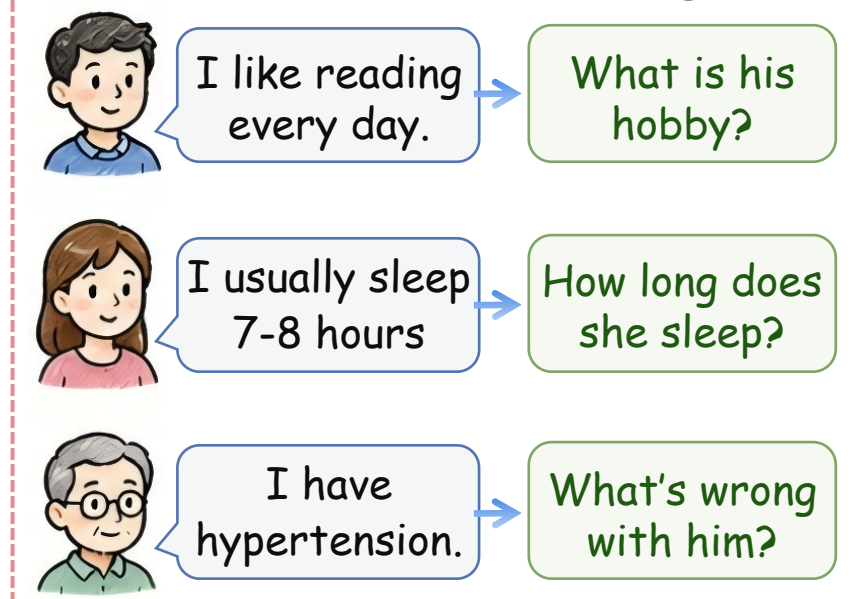
④ Noisy Memory Accumulation

Real Interactions



Existing Benchmarks

Clean / curated dialogue



Most dialogue facts appear in queries

Lack of saturation pressure